Mathematical Modeling and High-Precision Cascaded PID Controller Derivation for a 6-DOF Quadrotor

Autonomous Aerial Systems Course

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Abstract

This document presents the complete mathematical derivation of the 6-DOF rigid-body dynamics for an X-configuration quadrotor in MuJoCo, along with the step-by-step synthesis of a high-precision cascaded position, heading-decoupled attitude, and altitude PID controller. The analytical inverse mixer allocation matrix and tilt compensation laws ensuring zero steady-state error and asymptotic convergence are derived in detail.

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1 Coordinate Systems and Kinematics

We define two primary coordinate frames:

1. **Inertial/World Frame** $\mathcal{F}_W = \{\mathbf{x}_W, \mathbf{y}_W, \mathbf{z}_W\}$: An Earth-fixed Cartesian coordinate system with $\mathbf{z}_W$ pointing upward against gravity.
2. **Body-Fixed Frame** $\mathcal{F}_B = \{\mathbf{x}_B, \mathbf{y}_B, \mathbf{z}_B\}$: Attached to the quadrotor's center of mass (CoM), with $\mathbf{x}_B$ pointing forward, $\mathbf{y}_B$ pointing left/right, and $\mathbf{z}_B$ pointing upward perpendicular to the propeller plane.

1.1 State Representation

The complete 12-dimensional state vector $\mathbf{x} \in \mathbb{R}^{12}$ is given by:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p}^T & \mathbf{v}^T & \boldsymbol{\eta}^T & \boldsymbol{\omega}^T \end{bmatrix}^T \tag{1}$$

where:

- $\mathbf{p} = [x, y, z]^T \in \mathbb{R}^3$: Position of the quadrotor CoM in the world frame $\mathcal{F}_W$.
- $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^T \in \mathbb{R}^3$: Linear velocity in $\mathcal{F}_W$.
- $\boldsymbol{\eta} = [\phi, \theta, \psi]^T \in \mathbb{R}^3$: Euler angles representing Roll, Pitch, and Yaw.
- $\boldsymbol{\omega} = [p, q, r]^T \in \mathbb{R}^3$: Angular velocity vector expressed in the body-fixed frame $\mathcal{F}_B$.

1.2 Attitude Kinematics

The rotation matrix $\mathbf{R} \in SO(3)$ that transforms vectors from the body frame $\mathcal{F}_B$ to the world frame $\mathcal{F}_W$ ($\mathbf{v}_W = \mathbf{R}\mathbf{v}_B$) is constructed using the standard Z-Y-X (Yaw-Pitch-Roll) Tait-Bryan convention:

$$\mathbf{R}(\phi, \theta, \psi) = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi) \tag{2}$$

Expanding the elementary rotation matrices:

$$\mathbf{R} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix} \tag{3}$$

For small angular deviations ($\phi \approx 0, \theta \approx 0$), the transformation between body angular rates $\boldsymbol{\omega}$ and Euler angle derivatives $\dot{\boldsymbol{\eta}}$ simplifies to:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \approx \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{4}$$

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2 6-DOF Rigid-Body Dynamics

The motion of the quadrotor is governed by the Newton-Euler equations for a 6-DOF rigid body.

2.1 Translational Dynamics

Applying Newton's second law in the inertial frame $\mathcal{F}_W$:

$$m\ddot{\mathbf{p}} = \mathbf{R}\mathbf{F}_B + \mathbf{F}_g \tag{5}$$

where m is the total quadrotor mass ($m = 0.500$ kg), $\mathbf{F}_g = [0, 0, -mg]^T$ is the gravitational force, and $\mathbf{F}_B = [0, 0, T]^T$ is the total collective thrust produced along the body $\mathbf{z}_B$ axis.

Writing the equations component-wise:

$$m\ddot{x} = T(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \tag{6}$$

$$m\ddot{y} = T(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \tag{7}$$

$$m\ddot{z} = T(\cos\phi\cos\theta) - mg \tag{8}$$

2.2 Rotational Dynamics

Applying Euler's rotational equations in the body-fixed frame $\mathcal{F}_B$:

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \boldsymbol{\tau}_B \tag{9}$$

where $\mathbf{I} = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal moment of inertia tensor, and $\boldsymbol{\tau}_B = [\tau_x, \tau_y, \tau_z]^T$ is the vector of external control torques in the body frame.

Expanding component-wise:

$$I_{xx}\dot{p} = \tau_x - (I_{zz} - I_{yy})qr \tag{10}$$

$$I_{yy}\dot{q} = \tau_y - (I_{xx} - I_{zz})pr \tag{11}$$

$$I_{zz}\dot{r} = \tau_z - (I_{yy} - I_{xx})pq \tag{12}$$

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3 Rotor Thrust Allocation and Analytical Mixer Matrix

Consider an X-configuration quadrotor where the 4 rotors are positioned at a distance L from the center of mass along 45° diagonals ($d = 0.12$ m):

- **Rotor 0 (Front-Right)**: $\mathbf{r}_0 = [+d, +d, 0]^T$
- **Rotor 1 (Rear-Right)**: $\mathbf{r}_1 = [-d, +d, 0]^T$
- **Rotor 2 (Rear-Left)**: $\mathbf{r}_2 = [-d, -d, 0]^T$
- **Rotor 3 (Front-Left)**: $\mathbf{r}_3 = [+d, -d, 0]^T$

3.1 Torque Generation

Each rotor produces a thrust force $\mathbf{F}_i = [0, 0, T_i]^T$ and an aerodynamic reaction drag torque $\mathbf{M}_{d,i} = [0, 0, \pm c_\tau T_i]^T$ opposite to its spin direction ($c_\tau = 0.015$ m).

The total torque generated by the rotors about the CoM is:

$$\boldsymbol{\tau}_B = \sum_{i=0}^{3} (\mathbf{r}_i \times \mathbf{F}_i + \mathbf{M}_{d,i}) \tag{13}$$

Computing the cross products $\mathbf{r}_i \times \mathbf{F}_i$:

$$\mathbf{r}_0 \times \mathbf{F}_0 = [+d, +d, 0]^T \times [0, 0, T_0]^T = [+dT_0, -dT_0, 0]^T \tag{14}$$

$$\mathbf{r}_1 \times \mathbf{F}_1 = [-d, +d, 0]^T \times [0, 0, T_1]^T = [+dT_1, +dT_1, 0]^T \tag{15}$$

$$\mathbf{r}_2 \times \mathbf{F}_2 = [-d, -d, 0]^T \times [0, 0, T_2]^T = [-dT_2, +dT_2, 0]^T \tag{16}$$

$$\mathbf{r}_3 \times \mathbf{F}_3 = [+d, -d, 0]^T \times [0, 0, T_3]^T = [-dT_3, -dT_3, 0]^T \tag{17}$$

3.2 Forward Allocation Matrix

Summing the thrust and torque components yields the linear mapping:

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ d & d & -d & -d \\ -d & d & d & -d \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix} \begin{bmatrix} T_0 \\ T_1 \\ T_2 \\ T_3 \end{bmatrix} \tag{18}$$

3.3 Exact Inverse Mixer Matrix

Inverting the 4×4 allocation matrix analytically yields the exact rotor command equations:

$$\boxed{\begin{aligned} T_0 &= \frac{1}{4}T + \frac{1}{4d}\tau_x - \frac{1}{4d}\tau_y + \frac{1}{4c_\tau}\tau_z \\ T_1 &= \frac{1}{4}T + \frac{1}{4d}\tau_x + \frac{1}{4d}\tau_y - \frac{1}{4c_\tau}\tau_z \\ T_2 &= \frac{1}{4}T - \frac{1}{4d}\tau_x + \frac{1}{4d}\tau_y + \frac{1}{4c_\tau}\tau_z \\ T_3 &= \frac{1}{4}T - \frac{1}{4d}\tau_x - \frac{1}{4d}\tau_y - \frac{1}{4c_\tau}\tau_z \end{aligned}} \tag{19}$$

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4 High-Precision Cascaded PID Controller Synthesis

Quadrotors are underactuated systems with 4 control inputs and 6 degrees of freedom. We implement a **cascaded position, heading-decoupled attitude, and altitude PID controller**.

4.1 Altitude PID Control with Tilt Compensation

Let $e_z = z_{\text{target}} - z$. The desired vertical acceleration with integral error $I_z = \int e_z \, dt$ is:

$$a_{z,\text{cmd}} = k_{p,z} e_z - k_{d,z}\dot{z} + k_{i,z} I_z \tag{20}$$

To maintain altitude during banking and pitch maneuvers, the collective thrust includes **tilt compensation**:

$$T = \frac{m(g + a_{z,\text{cmd}})}{\max(0.65,\ \cos\phi\cos\theta)} \tag{21}$$

4.2 Horizontal Position PID and Yaw Decoupling

Let $e_x = x_{\text{target}} - x$ and $e_y = y_{\text{target}} - y$. Desired world-frame accelerations are:

$$a_{x,\text{world}} = k_{p,xy} e_x - k_{d,xy}\dot{x} + k_{i,xy} \int e_x \, dt \tag{22}$$

$$a_{y,\text{world}} = k_{p,xy} e_y - k_{d,xy}\dot{y} + k_{i,xy} \int e_y \, dt \tag{23}$$

To ensure consistent control authority regardless of the quadrotor's current yaw angle ψ, we rotate world accelerations into the quadrotor's **heading frame**:

$$\begin{bmatrix} a_{x,\text{body}} \\ a_{y,\text{body}} \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi \\ -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} a_{x,\text{world}} \\ a_{y,\text{world}} \end{bmatrix} \tag{24}$$

The desired roll and pitch tilt angles are given by:

$$\theta_{\text{des}} = \text{clip}\left(\frac{a_{x,\text{body}}}{g}, -\theta_{\max}, \theta_{\max}\right), \quad \phi_{\text{des}} = \text{clip}\left(-\frac{a_{y,\text{body}}}{g}, -\phi_{\max}, \phi_{\max}\right) \tag{25}$$

4.3 Attitude and Angular Rate PD Control

The inner loop regulates the Euler angles to $(\phi_{\text{des}}, \theta_{\text{des}}, \psi_{\text{des}})$ by commanding control torques:

$$\tau_x = k_{p,\phi}(\phi_{\text{des}} - \phi) - k_{d,\phi} p \tag{26}$$

$$\tau_y = k_{p,\theta}(\theta_{\text{des}} - \theta) - k_{d,\theta} q \tag{27}$$

$$\tau_z = k_{p,\psi}(\psi_{\text{des}} - \psi) - k_{d,\psi} r \tag{28}$$

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5 Summary of Parameters

Table 1 lists the physical and control parameters used in the MuJoCo simulation environment:

Table 1: Quadrotor Physical and Controller Parameters

Parameter	**Symbol**	**Value**	**Unit**
Total Mass	m	0.500	kg
Arm Half-Length	d	0.12	m
Gravitational Acceleration	g	9.81	m/s^2
Moments of Inertia	I_{xx}, I_{yy}, I_{zz}	$0.0025, 0.0025, 0.0045$	$\text{kg} \cdot \text{m}^2$
Torque-to-Thrust Ratio	c_τ	0.015	m
Altitude Gains	$k_{p,z}, k_{d,z}, k_{i,z}$	$12.0, 6.0, 4.0$	$\text{s}^{-2}, \text{s}^{-1}, \text{s}^{-3}$
Position Gains	$k_{p,xy}, k_{d,xy}, k_{i,xy}$	$1.8, 2.0, 0.5$	$\text{s}^{-2}, \text{s}^{-1}, \text{s}^{-3}$
Attitude Proportional Gains	$k_{p,\phi}, k_{p,\theta}, k_{p,\psi}$	$0.075, 0.075, 0.020$	$\text{N} \cdot \text{m/rad}$
Attitude Derivative Gains	$k_{d,\phi}, k_{d,\theta}, k_{d,\psi}$	$0.015, 0.015, 0.005$	$\text{N} \cdot \text{m} \cdot \text{s/rad}$
Max Commanded Tilt	$\phi_{\max}, \theta_{\max}$	0.22	rad ($\approx 12.6°$)